

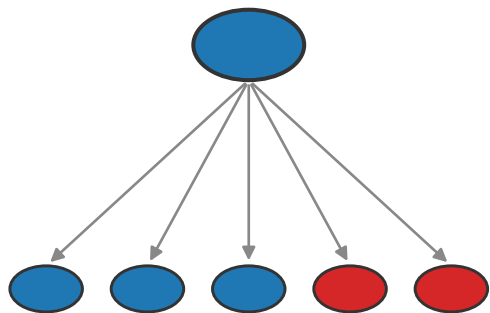
 prefers block A

 prefers block B

$K = 5$ sampled each round (not the whole network)

round 1

validator v



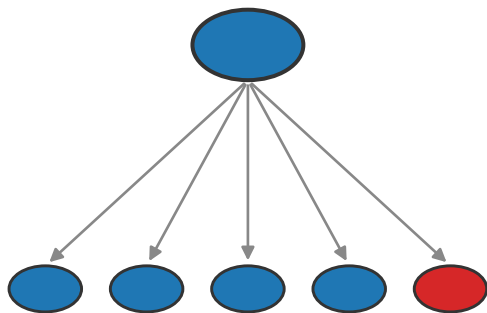
query K random peers

sample: 3/5 blue
counter $\rightarrow 1$



round 2

validator v



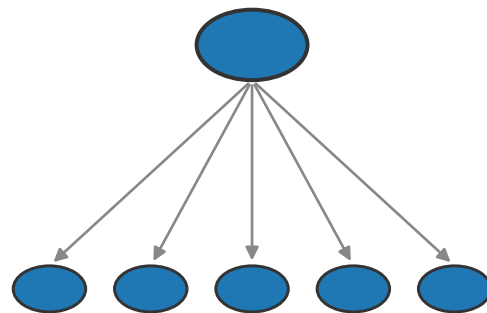
query K random peers

sample: 4/5 blue
counter $\rightarrow 2$



...

validator v



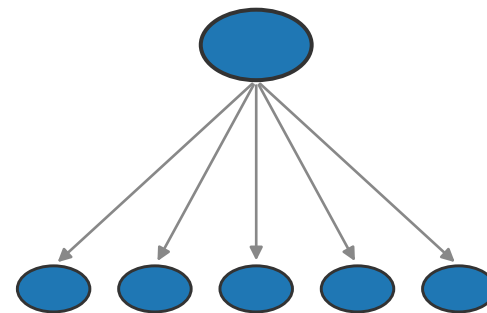
query K random peers

sample: 5/5 blue
counter $\rightarrow \dots$



round β

validator v



query K random peers

sample: 5/5 blue
counter $\rightarrow \beta$

counter reaches $\beta \rightarrow$ **ACCEPT**
probabilistic finality, $\epsilon \leq (1 - \alpha_c/K)^\beta$